Precambrian Research, 3 (1976) 231—241

© Elsevier Scientific Publishing Company, Amsterdam — Printed in The Netherlands

PALAEOMAGNETIC RESULTS FROM THE HART DOLERITE OF THE KIMBERLEY BLOCK, AUSTRALIA

M.W. McELHINNY and M.E. EVANS*

Research School of Earth Sciences, Australian National University, Canberra, A.C.T. (Australia)

(Received March 11, 1975; revision accepted June 13, 1975)

ABSTRACT

McElhinny, M.W. and Evans, M.E., 1976. Palaeomagnetic results from the Hart dolerite of the Kimberley Block, Australia. Precambrian Res., 3: 231—241.

A routine palaeomagnetic study of the 1800 Ma old Hart dolerite shows that sample directions of magnetism give good within-site grouping at 12 sites after magnetic cleaning. However, the site mean directions are scattered over one-quarter of the sphere, a feature that apparently does not arise from poor structural control. A baked contact test confirms that the magnetization is original. Comparison of bulk NRM, TRM, ARM and saturation IRM properties has been made with the Modipe gabbro of southern Africa from which a palaeointensity has previously been estimated. These measurements are consistent with the hypothesis that the main dipole field intensity temporarily was much reduced with respect to the non-dipole field 1800 m.y. ago resulting in increased directional scatter from secular variation. The average of the site VGPs at 29N 46E with $A_{95} = 24°$ thus is still a viable estimate of the palaeomagnetic pole at that time bearing in mind the larger than usual statistical error associated with it.

GEOLOGY AND SAMPLING

In the northwestern part of Western Australia lies the Kimberley Block, a large structural basin covering the whole of the Kimberley Plateau area and bounded on the southwest and southeast by rocks of the Halls Creek Mobile Belt. The basal unit in the Kimberley Basin is the Whitewater volcanics which has an age of 1823 ± 17 Ma from an Rb—Sr total rock isochron on ten samples using decay constant $1.39 \cdot 10^{-11}$ yr^{-1} (Bofinger, 1967). They are overlain uncomformably by the Speewah and Kimberley Groups which are themselves conformable. The Carson Volcanics, a unit within the Kimberley Group has an age of 1807 Ma from Rb—Sr total rock measurements on two samples. The sequence is intruded extensively by sills of the Hart dolerite which has

*Present address: Institute of Earth and Planetary Physics, University of Alberta, Edmonton, Alta., Canada.

an Rb—Sr total rock isochron age of 1800 ± 25 Ma determined from nine samples (Bofinger, 1967). The Proterozoic of Australia has been divided into three Systems which from oldest to youngest have been named Nullaginian, Carpentarian and Adelaidean (Dunn et al., 1966). In terms of this subdivision of the Precambrian time-scale in Australia the age of the Hart dolerite corresponds to the lower boundary of the Carpentarian System (McDougall et al., 1965; Dunn et al., 1966).

The Hart dolerite is exposed over a total area of approximately 4700 km^2. Detailed descriptions of its occurrence and petrography have been made by Gellatty et al. (1965) who describe it as the most extensively developed Precambrian dolerite known, the total area of outcrop being of the same order as that of the Tasmanian dolerite. It consists of a series of anastomosing sills with associated granophyres, the thickness of individual sills ranging from 20 to 2000 m, the largest intruding the dominantly arenaceous Speewah Group. Its great extent, relatively good exposure, and excellent geochronological control make it especially suitable for palaeomagnetic investigations.

Three hand samples were collected from each of 18 sites (Fig. 1), including some baked sediments adjacent to in situ dolerite. Samples were oriented magnetically, but as the dolerite is weakly magnetized, this technique is considered adequate. Although the occurrence of the dolerite is widespread, sampling was restricted to the region shown in Fig. 1, where there are a number of excellent river and creek bed sections with relatively easy access. The sampling sites were distributed mainly along the strike, but sites A to G represent a traverse across the outcrop of the thickest sill. Samples were also

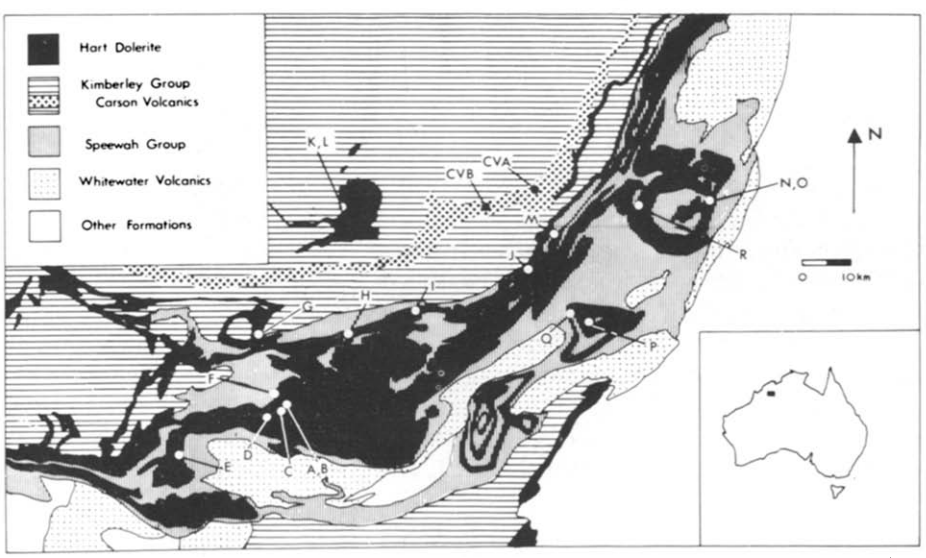


Fig. 1. Sampling localities in the Hart dolerite (A to R) and the Carson volcanics (CVA and CVB).

collected from two sites within the Carson Volcanics (CVA and CVB in Fig. 1), but exposures of this unit are generally poor and the volcanics somewhat altered. No useful palaeomagnetic information could be obtained from the Carson Volcanics and they are not discussed further.

PALAEOMAGNETIC MEASUREMENTS

Test specimens from each sample were subjected to increasing alternating magnetic fields and the Briden (1972) optimum cleaning index and stability index of Giddings and McElhinny (1975) were determined. The stability indices (Fig. 2) indicate that the overall palaeomagnetic stability of these rocks was not good. However, the improvement in grouping during demagnetization was usually very marked and the end-points of the demagnetizing procedure well determined as shown by the values of Briden's (1972) index (Fig. 2b). The range of stable fields was limited and most specimen directions tended to diverge again after only two or three demagnetizing steps. This feature results in overall low stability classification. Only sites K and L exhibit systematic movement during demagnetization followed by tightly grouped stable end-points. These two sites were collected from a small sill of very fresh dolerite, K, and siltstone of the Teronis member of the Kimberley Group, L. The siltstone site is estimated to lie within 50 cm of the contact and was therefore almost certainly baked when the sill was intruded. The agreement between the mean directions of magnetization at these sites strongly suggests this to be the case (Table I), although there is very little indication of this in

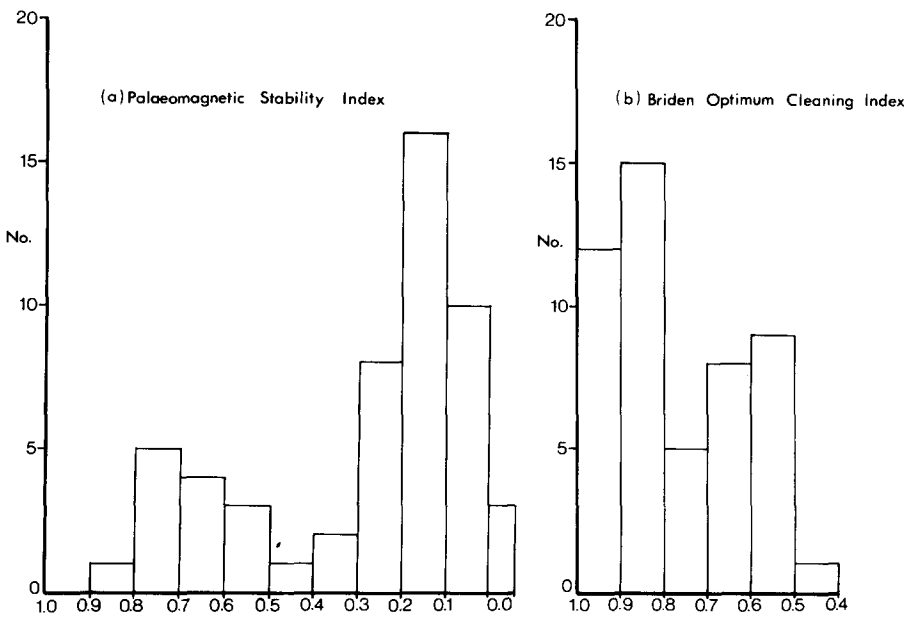


Fig. 2. Histogram of stability indices observed in specimens of the Hart dolerite.

TABLE I

Mean directions from 12 sites that responded successfully to magnetic cleaning in the peak field B indicated; three samples were collected at each site and the mean site position lies at 17.5S, 127.1E

Site	B(mT)	D	I	R	k	Site	VGP
A	20	151	+ 3	2.980	100	57N	63E
B	5	114	−57	2.629	5	7N	80E
C	10	209	−25	2.921	25	48N	172E
D	30	300	−51	2.948	39	34N	10E
I	20	83	+53	2.967	61	4N	2E
K	20	120	+15	2.977	87	31N	38E
L	40	119	+ 4	2.975	79	28N	44E
M	10	77	−48	2.974	76	20N	63E
N	15	313	−47	2.965	57	46N	15E
O	10	143	−25	2.945	37	42N	76E
P	15	97	+10	2.859	14	8N	36E
Q	15	92	− 9	2.873	16	1N	42E

thin sections, apart from suturing of quartz grain boundaries which may represent the beginning of recrystallization.

After magnetic cleaning only 12 of the 18 sites gave consistent directions of magnetization as listed in Table I and illustrated in Fig. 3. The site means have fairly widely scattered directions which cover nearly a quarter of the sphere. Assuming that sites D and N have opposite polarity to the other 10 sites, the 12 site means are analysed in various ways in Table II. Tectonic corrections (generally about 10° but ranging up to 27°) determined from the dip of nearby sediments have been applied to each site mean. Without these corrections the overall mean direction is $D = 102°$, $I = 7°$ with $k = 2.74$, so that the grouping is slightly, but not significantly, better after the dip correction is made. The palaeomagnetic pole position is best obtained from the mean of the site poles and lies at 29N, 46E with $A_{95} = 24°$ when all 12 sites are considered. Eliminating the most deviant sites B and C makes little difference to the overall mean or pole position. Even if only the two most stable sites K and its baked contact L are used, the mean changes very little. The large scatter is puzzling, however, especially for fresh dolerite material of this kind.

To investigate further the source of scatter, pilot thermal demagnetization was carried out on one specimen from each of ten of the sites that responded to magnetic cleaning. Heating and cooling cycles were carried out in a nitrogen atmosphere. Although poorly grouped, the overall mean direction of these pilot specimens remains practically unchanged between 200 and 550°C (Table III), the individual directions (Fig. 4) agreeing reasonably well with the corresponding site means obtained after magnetic cleaning. The thermal demagnetization curves show intensity decreases of one-third to one-half up

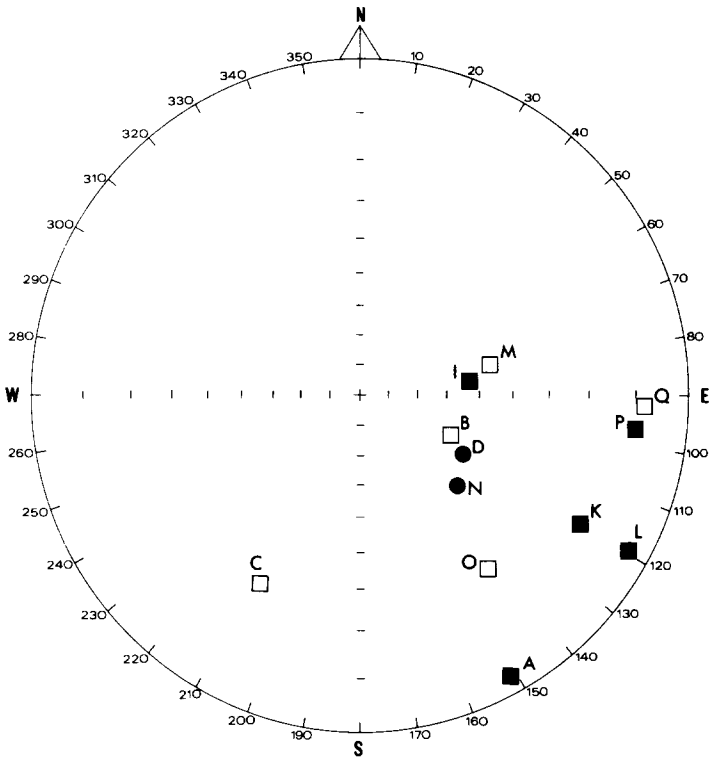


Fig. 3. Mean directions at 12 sites in the Hart dolerite after magnetic cleaning. Stereographic projection with squares representing north-seeking directions on the lower (solid squares) and upper (open squares) hemisphere. Solid circles represent south-seeking directions on the lower hemisphere and are regarded as having the opposite polarity.

to 550°C after which there is a sharp decrease near the Curie point of magnetite. Thermal demagnetization therefore confirms the scatter observed from magnetic cleaning but also shows that the specimens have the characteristics of moderately stable material. The low stability under alternating field demagnetization is unusual. It seems unlikely that the scatter can be due to structural complications because the region has not been subjected to intense deformation, the tectonic corrections being relatively small. It appears that the scatter in site mean directions may be a feature of the geomagnetic field at the time of intrusion of the dolerite. This possibility is investigated further in the next section.

EVIDENCE FOR REDUCED FIELD INTENSITY

It is possible that the scattered directions can be explained by polar wandering during the intrusion of the dolerite, such as was observed by Hargraves

TABLE II

Formation mean of site directions and VGPs calculated in different ways

Sites used	N	Site directions				
		D	I	R	k	α_{95}
ABCDIKLMNOPQ	12	121	+ 2	8.260	2.94	31
A—CDIKLMNOPQ	11	122	+ 8	7.796	3.12	31
AB—DIKLMNOPQ	11	115	+ 5	8.299	3.70	28
A——DIKLMNOPQ	10	115	+11	7.876	4.24	27
KL	2	119	+10	1.991	108	24
		Site poles				
		lat. N	long. E	R	K	A_{95}
ABCDIKLMNOPQ	12	29	46	9.360	4.17	24
A—CDIKLMNOPQ	11	31	42	8.605	4.18	25
AB—DIKLMNOPQ	11	24	42	9.397	6.24	20
A——DIKLMNOPQ	10	25	38	8.654	6.69	20
KL	2	30	42	1.998	452	12

TABLE III

Thermal demagnetization of ten test specimens

Temp. (°C)	D	I	k	α_{95}
200	129	+20	3.6	29
300	123	+18	3.3	31
400	122	+14	3.6	29
450	119	+10	3.3	31
500	119	+ 8	3.3	31
550	125	+ 0	3.8	28

(1968) in the Roraima dolerites of South America. However, the scatter here is observed as much in the single thick sill as between sills. On the basis of the cooling time polar wandering seems unlikely since this sill would have cooled to 600°C at its centre after about $0.02D^2$ yr $\approx 2 \cdot 10^5$ yr (Jaeger, 1957) where D is the sill thickness in meters. Another possibility is that the scatter of directions is due to increased secular variation during the period in which the dolerite cooled brought about by temporary reduction in strength of the main dipole field relative to the non-dipole field.

The circular standard deviation of a Fisherian distribution, $\delta = 81/k^{1/2}{}^\circ$ for directions or $\Delta = 81/K^{1/2}{}^\circ$ for the virtual poles is usually used as a measure of

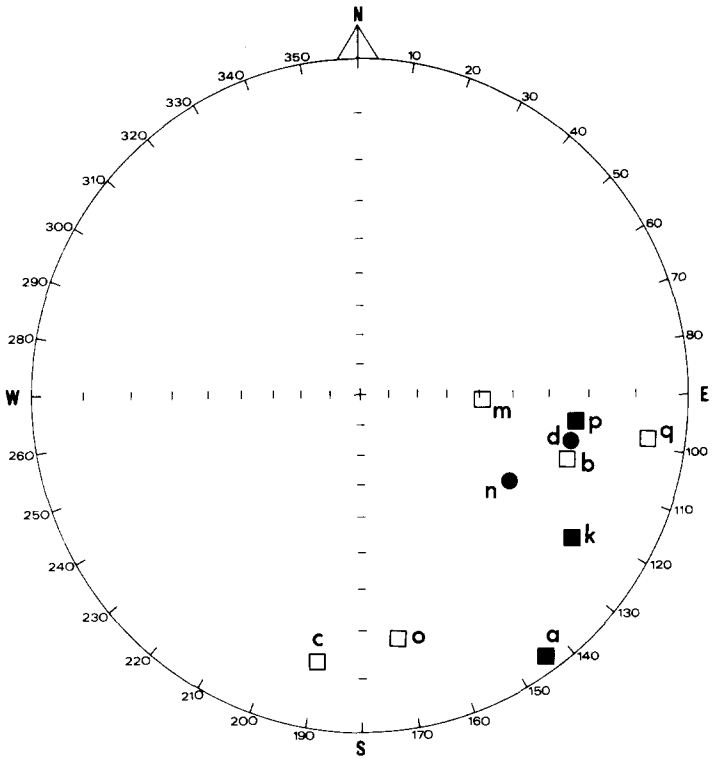


Fig. 4. Test specimens from ten of the sites in Fig. 3 after thermal cleaning at 550°C. Projection and symbols as in Fig. 3.

palaeosecular variation. In the present case the 12 site means give $\delta = 47°$ and $\Delta = 40°$ for a palaeolatitude of only 1°. This is very much higher than would be expected from various models of palaeosecular variation and from average palaeosecular variation observed in palaeomagnetic results (Creer, 1962; Irving and Ward, 1964; Cox, 1970; Brock, 1971; McElhinny and Merrill, 1975). Using the most recent analysis of McElhinny and Merrill (1975) the average dispersion of virtual poles observed over the last 5 million years from palaeomagnetic results is 13.2° at the equator. The value deduced from the Hart dolerite is three times higher. The total angular dispersion S_T is given by

$$S_T^2 = S_D^2 + S_N^2$$

where S_D is the dispersion due to dipole wobble and S_N is the dispersion due to the non-dipole field. At the equator

$$S_N = a(f_N/F)$$

where a is constant, f_N is the non-dipole field intensity, and F is the dipole field intensity.

Taking $S_D = 9°$ as estimated by McElhinny and Merrill (1975) for the past

5 million years, a value of S_N of 39° is obtained for the Hart dolerite. This compares with $S_N = 9.7°$ over the past 5 million years. If the non-dipole field remains unchanged the requirement is that the equatorial field strength be reduced by a factor of four.

To investigate the possibility that the palaeointensity was low during the intrusion of the Hart dolerite we have compared intensity of magnetizations of NRM and TRM with that observed in the Modipe gabbro of Botswana (Evans and McElhinny, 1966) for which a palaeointensity determination has already been made (McElhinny and Evans, 1968). These are both old Precambrian intrusions, the age of the Modipe gabbro being about 2700 m.y. Both have low $^{87}Sr/^{86}Sr$ initial ratios of 0.7010 for Modipe (McElhinny, 1966) and 0.7026 for the Hart (Bofinger, 1967) indicating they are most probably uncontaminated mantle material. The ratio NRM/TRM may not necessarily provide a realistic estimate of the palaeointensity for either formation by itself, but a comparison of the ratios for the two formations enables the *relative* palaeointensities to be estimated. Secondary magnetizations have had to be removed from the NRM through alternating field demagnetization and in most cases this was complete at about 10 mT. Therefore we have compared the intensities of NRM and TRM after demagnetization in a field of 10 mT in each case. If F_e is the equatorial field intensity and λ_H, λ_M are the palaeolatitudes for the Hart dolerite and Modipe gabbro respectively, then:

$$\frac{\left(\frac{NRM}{TRM}\right)_{Hart}}{\left(\frac{NRM}{TRM}\right)_{Modipe}} = \frac{(F_e)_{Hart}\,(1 + 3\sin^2\lambda_H)^{1/2}}{(F_e)_{Modipe}\,(1 + 3\sin^2\lambda_M)^{1/2}}$$

$$\approx \frac{(F_e)_{Hart}}{3\,(F_e)_{present}}$$

since $\lambda_H \approx 0°$, $\lambda_M \approx 80°$ and $(F_e)_{Modipe} \approx 1.5\,(F_e)_{present}$ from the palaeointensity study of McElhinny and Evans (1968). Using only those specimens which are magnetically cleaned by 10 mT alternating magnetic field geometric mean intensities have been calculated for the two rock units for their NRM and TRM (after demagnetization in 10 mT) (Table IV). It is at once noted that whereas the mean intensity of NRM of the Hart dolerite is only about one-half that of the Modipe gabbro, its TRM is nearly ten times greater. Substituting these values in the equation above we obtain:

$$(F_e)_{Hart} \approx 0.15\,(F_e)_{present}$$

These values are consistent with the hypothesis that the dipole field strength was much reduced during the cooling of the Hart dolerite.

However, it can be argued that heating the specimens, especially the Hart

TABLE IV

Geometric mean intensities with standard error for NRM and TRM of N specimens (after demagnetization in 10 mT) from the Hart dolerite and Modipe gabbro

	N	Hart dolerite	N	Modipe gabbro
NRM	42	$0.096^{+0.018}_{-0.015}$ $A\ m^{-1}$	32	$0.216^{+0.061}_{-0.048}$ $A\ m^{-1}$
TRM (0.06 mT)*	30	$8.64^{+0.92}_{-0.88}$ $A\ m^{-1}$	36	$0.94^{+0.15}_{-0.12}$ $A\ m^{-1}$

*Some samples of Modipe gabbro were given TRM in fields of 0.03 or 0.1 mT. Linear relationship between TRM and applied field was assumed.

dolerite, in order to give them a TRM has produced such magneto-chemical changes that their ability to acquire TRM has been strongly enhanced. To check on this possibility 20 specimens of Hart dolerite and 19 of the Modipe gabbro were each subjected to a series of remanence acquisition tests before and after heating to 600°C (Table V). All specimens were first given an ARM in 0.1 mT steady field using an alternating magnetic field of 200 mT followed by a saturation IRM (SIRM) in a field of 0.55 T. They were then heated to 600°C in a nitrogen atmosphere and acquired a TRM on cooling in 0.1 mT. ARM and SIRM values were then again measured. The remarkable thing

TABLE V

Alteration checks on test specimens from the Hart dolerite (N=20) and Modipe gabbro (N=19) before (subscript b) and after (subscript a) heating to 600°C and cooling in a field of 0.1 mT; ARM = anhysteretic remanent magnetization; SIRM = saturation isothermal remanent magnetization; geometric mean intensities and standard error are given in units of $A\ m^{-1}$

	Hart dolerite	Modipe gabbro
ARM_b (0.1 mT)	$3.45^{+0.49}_{-0.43}$	$0.62^{+0.11}_{-0.93}$
ARM_a (0.1 mT)	$3.85^{+0.48}_{-0.42}$	$0.68^{+0.11}_{-0.98}$
ARM_a/ARM_b	1.11	1.10
$SIRM_b$ (0.55 T)	336^{+64}_{-54}	$13.2^{+4.3}_{-3.2}$
$SIRM_a$ (0.55 T)	456^{+67}_{-58}	$17.4^{+4.7}_{-3.6}$
$SIRM_a/SIRM_b$	1.36	1.33
TRM (0.1 mT)	$18.1^{+2.5}_{-2.2}$	$1.85^{+0.42}_{-0.34}$
ARM_a/TRM	0.21	0.37

about the values given in Table V is that the changes in both suites of specimens are virtually identical; ARM values increase by about 10% on heating and SIRM values by 35%. Since ARM is the closer analogue of low field TRM the changes are minimal. In any case the effect of such changes operates in the same sense for both suites of rocks. ARM/TRM ratios for both suites are sufficiently less than one that complications due to the presence of multi-domain grains are probably excluded (Levi, 1973). Thus the observation that the Hart dolerite acquires an ARM about five times greater than the Modipe gabbro supports the palaeointensity comparison.

CONCLUSIONS

The palaeointensity comparisons with the Modipe gabbro suggest that the main dipole field was considerably reduced during the period the Hart dolerite cooled. This can account for the increased scatter as being due to an enhancement of the non-dipole field with respect to the main dipole field. However, weak geomagnetic field strengths commonly occur during polarity transitions. It might be argued therefore that the grouping of stable directions reflects a transitional situation in which the direction pauses for a while half-way through a transition (Dagley and Lawley, 1974). Polarity transitions are estimated to take about $2 \cdot 10^3$ yr to complete (McElhinny, 1971) whereas we have estimated the palaeomagnetic directions cover a time span at least 100 times longer from the cooling time of the large sill. It seems very unlikely therefore that we have fortuitously sampled the central part of a polarity transition. In view of this we believe the pole position at 29N, 46E still represents a realistic estimate of the palaeomagnetic pole position at 1800 Ma, the large error circle ($A_{95} = 24^\circ$) properly reflecting the uncertainty arising from relatively large secular variation at the time. Standard palaeomagnetic tests have confirmed that the cleaned remanence is probably primary.

ACKNOWLEDGEMENTS

We thank K.A. Plumb of the Bureau of Mineral Resources who advised us on the most suitable sampling sites in the East Kimberley region. D. Edwards kindly assisted with some of the magnetic measurements, and R. Merrill provided valuable discussion on the alteration checks.

REFERENCES

Bofinger, V.M., 1967. Geochronology of the East Kimberley area of Western Australia. Ph.D. Thesis, Australian National University.

Briden, J.C., 1972. A stability index of remanent magnetism. J. Geophys. Res., 77: 1401—1405.

Brock, A., 1971. An experimental study of palaeosecular variation. Geophys. J. R. Astron. Soc., 24: 303—317.

Cox, A., 1970. Latitude dependence of the angular dispersion of the geomagnetic field. Geophys. J. Roy. Astron. Soc., 20: 253—269.
Creer, K.M., 1962. The dispersion of the geomagnetic field due to secular variation and its determination for remote times from palaeomagnetic data. J. Geophys. Res., 67: 3461.
Dagley, P. and Lawley, E., 1974. Palaeomagnetic evidence for the transitional behaviour of the geomagnetic field. Geophys. J. R. Astron. Soc., 36: 577, 598.
Dunn, P.R., Plumb, K.A. and Roberts, H.G., 1966. A proposal for time-stratigraphic subdivision of the Australian Precambrian. J. Geol. Soc. Aust., 13: 593—608.
Evans, M.E. and McElhinny, M.W., 1966. The Palaeomagnetism of the Modipe gabbro. J. Geophys. Res., 71: 6053—6063.
Gellatty, D.C., Derrick, G.M. and Plumb, K.A., 1965. The geology of the Landsdowne 1:250.000 sheet Se 52/5 Western Australia. Bur. Min. Resour. Aust. Rec. No. 210.
Giddings, J.G. and McElhinny, M.W., 1975. A new index of palaeomagnetic stability for magnetite bearing igneous rocks. Geophys. J. Roy. Astron. Soc. (in press).
Hargraves, R.B., 1968. Palaeomagnetism of the Roraima dolerites. Geophys. J.R. Astron. Soc., 16: 147—160.
Irving, E. and Ward, M.A., 1964. A statistical model of the geomagnetic field. Pure Appl. Geophys., 57: 47.
Jaeger, J.C., 1957. The temperature in the neighbourhood of a coiling intrusive sheet. Am. J. Sci., 255: 306—318.
Levi, S., 1973. Comparison of some ARM and TRM properties in magnetites. Trans. Am. Geophys. Union, 54: 1071 (abstract).
McDougall, I., Dunn, P.R., Compston, W., Webb, A.M., Richards, J.R. and Bofinger, V.M., 1965. Isotopic age determinations on Precambrian rocks of the Carpentaria region, Northern Territory, Australia. J. Geol. Soc. Aust., 12: 67—90.
McElhinny, M.W., 1966. Rb—Sr and K—Ar measurements on the Modipe gabbro of Bechuanaland and South Africa. Earth Planet. Sci. Lett., 1: 439—442.
McElhinny, M.W., 1971. Geomagnetic polarity transitions. Comments on Earth Sci. Geophys., 1: 150—158.
McElhinny, M.W. and Evans, M.E., 1968. An investigation of the strength of the geomagnetic field in the early Precambrian. Phys. Earth Planet. Inter., 1: 485—497.
McElhinny, M.W. and Merrill, R.T., 1975. Geomagnetic secular variation over the past 5 million years. Rev. Geophys. Space Phys. (in press).